A team is developing a updation features with the following tasks: Requirement Gathering (2 days), Backend Design (4 days, after requirements), Frontend Design (4 days, after requirements), Backend Development (8 days, after backend design), Frontend Development (7 days, after frontend design), Testing (5 days, after both developments), and Release (2 days, after testing). Backend and Frontend Design can occur in parallel, as can Backend and Frontend Development.

a) Create a Gantt chart to illustrate the project timeline.

b) calculate the expected duration using O, M, and P estimates.

c) Determine the critical path of the project.

```
Day:         1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21
-----------------------------------------------------------------
Req. Gath.   █ █
BE Design        █ █ █ █
FE Design        █ █ █ █
BE Dev                   █ █ █ █ █ █ █ █
FE Dev                   █ █ █ █ █ █ █
Testing                                      █ █ █ █ █
Release                                               █ █
```

A Gantt chart visually represents the project timeline, showing task durations and their start/end times, accounting for dependencies and parallel tasks. The tasks are:

- **Requirement Gathering**: 2 days
- **Backend Design**: 4 days (after Requirement Gathering)
- **Frontend Design**: 4 days (after Requirement Gathering)
- **Backend Development**: 8 days (after Backend Design)
- **Frontend Development**: 7 days (after Frontend Design)
- **Testing**: 5 days (after Backend and Frontend Development)
- **Release**: 2 days (after Testing)

Dependencies and Parallelism:

- Backend Design and Frontend Design can occur in parallel after Requirement Gathering.
- Backend Development and Frontend Development can occur in parallel after their respective designs.
- Testing requires both Backend and Frontend Development to be complete.
- Release follows Testing.

Timeline Calculation:

- **Day 1–2**: Requirement Gathering (2 days).
- **Day 3–6**: Backend Design and Frontend Design (4 days each, in parallel).
- **Day 7–14**: Backend Development (8 days, starts after Backend Design).
- **Day 7–13**: Frontend Development (7 days, starts after Frontend Design, in parallel with Backend Development).
- **Day 15–19**: Testing (5 days, starts after both developments, i.e., after day 14 when Backend Development finishes, as it's the longer of the two).
- **Day 20–21**: Release (2 days, after Testing).

Total duration: 21 days

PERT (Program Evaluation and Review Technique) diagram represents tasks as nodes or edges with dependencies, and we'll calculate the expected duration for each task using Optimistic (O), Most Likely (M), and Pessimistic (P) estimates. Since only single durations are provided, we'll estimate O and P based on reasonable assumptions, using M as the given duration.

Assumptions for O, M, P Estimates:

- **Optimistic (O)**: 80% of the most likely duration (M), rounded to nearest 0.5 day.
- **Most Likely (M)**: Given duration.
- **Pessimistic (P)**: 120% of the most likely duration, rounded to nearest 0.5 day.

Expected Duration Formula:

$$\text{Expected Duration (TE)} = \frac{O+4M+P}{6}$$

Task Estimates:

- **Requirement Gathering**: M = 2 days
    - O = 2 × 0.8 = 1.6 ≈ 1.5 days
    - P = 2 × 1.2 = 2.4 ≈ 2.5 days
    - TE = (1.5 + 4×2 + 2.5) / 6 = (1.5 + 8 + 2.5) / 6 = 12 / 6 = 2 days

- **Backend Design**: M = 4 days
  - O = 4 × 0.8 = 3.2 ≈ 3 days
  - P = 4 × 1.2 = 4.8 ≈ 5 days
  - TE = (3 + 4×4 + 5) / 6 = (3 + 16 + 5) / 6 = 24 / 6 = 4 days
- **Frontend Design**: M = 4 days
  - O = 4 × 0.8 = 3.2 ≈ 3 days
  - P = 4 × 1.2 = 4.8 ≈ 5 days
  - TE = (3 + 4×4 + 5) / 6 = 24 / 6 = 4 days
- **Backend Development**: M = 8 days
  - O = 8 × 0.8 = 6.4 ≈ 6.5 days
  - P = 8 × 1.2 = 9.6 ≈ 10 days
  - TE = (6.5 + 4×8 + 10) / 6 = (6.5 + 32 + 10) / 6 = 48.5 / 6 ≈ 8.08 days
- **Frontend Development**: M = 7 days
  - O = 7 × 0.8 = 5.6 ≈ 5.5 days
  - P = 7 × 1.2 = 8.4 ≈ 8.5 days
  - TE = (5.5 + 4×7 + 8.5) / 6 = (5.5 + 28 + 8.5) / 6 = 42 / 6 = 7 days
- **Testing**: M = 5 days
  - O = 5 × 0.8 = 4 ≈ 4 days
  - P = 5 × 1.2 = 6 ≈ 6 days
  - TE = (4 + 4×5 + 6) / 6 = (4 + 20 + 6) / 6 = 30 / 6 = 5 days
- **Release**: M = 2 days
  - O = 2 × 0.8 = 1.6 ≈ 1.5 days
  - P = 2 × 1.2 = 2.4 ≈ 2.5 days
  - TE = (1.5 + 4×2 + 2.5) / 6 = 12 / 6 = 2 days

The critical path is the longest sequence of tasks determining the minimum project duration. Using the expected durations from the PERT analysis:

Paths:

- **Path 1**: A → B → D → F → G = 2 + 4 + 8.08 + 5 + 2 = **21.08 days**
- **Path 2**: A → C → E → F → G = 2 + 4 + 7 + 5 + 2 = **20 days**

The **critical path** is **Requirement Gathering → Backend Design → Backend Development → Testing → Release**, with an expected duration of **21.08 days**. This path is critical because any delay in these tasks directly impacts the project's completion time.

A startup is developing a mobile application for task management, estimated at 15 KLOC. The project is expected to have stable requirements and a small, experienced team. Using the COCOMO model, calculate the effort, development time, and team size for all three development modes (Organic, Semi-Detached, and Embedded).

The Basic COCOMO model provides equations for effort and development time based on the project's size (in KLOC) and development mode. The formulas are:

- **Effort (E)** in person-months: $E = a \times (KLOC)^b$
- **Development Time (T)** in months: $T = c \times (E)^d$
- **Average Team Size**: $\text{Team Size} = \frac{E}{T}$

The constants a, b, c, and d depend on the development mode:

Mode	a	b	c	d
Organic	2.4	1.05	2.5	0.38
Semi-Detached	3.0	1.12	2.5	0.35
Embedded	3.6	1.20	2.5	0.32

1. Organic Mode

- **Effort**:

$$E = 2.4 \times (15)^{1.05}$$

First, compute $15^{1.05}$:

$$15^{1.05} \approx 15 \times 15^{0.05} \approx 15 \times 1.116 = 16.74$$

(Using $15^{0.05} \approx e^{0.05\times\ln 15} \approx e^{0.05\times2.708} \approx e^{0.1354} \approx 1.116$).

$$E = 2.4 \times 16.74 \approx 40.18 \text{ person-months}$$

- **Development Time**:

$$T = 2.5 \times (40.18)^{0.38}$$

Compute $40.18^{0.38}$:

$$40.18^{0.38} \approx e^{0.38\times\ln 40.18} \approx e^{0.38\times3.692} \approx e^{1.403} \approx 4.07$$

$$T = 2.5 \times 4.07 \approx 10.18 \text{ months}$$

- **Team Size**:

$$\text{Team Size} = \frac{40.18}{10.18} \approx 3.95 \approx 4 \text{ persons}$$

2. Semi-Detached Mode

- **Effort**:

$$E = 3.0 \times (15)^{1.12}$$

Compute $15^{1abandons.12}$:

$$15^{1.12} \approx 15 \times 15^{0.12} \approx 15 \times 1.302 = 19.53$$

(Using $15^{0.12} \approx e^{0.12 \times \ln 15} \approx e^{0.12 \times 2.708} \approx e^{0.325} \approx 1.302$).

$$E = 3.0 \times 19.53 \approx 58.59 \text{ person-months}$$

- **Development Time**:

$$T = 2.5 \times (58.59)^{0.35}$$

Compute $58.59^{0.35}$:

$$58.59^{0.35} \approx e^{0.35 \times \ln 58.59} \approx e^{0.35 \times 4.07} \approx e^{1.424} \approx 4.15$$

$$T = 2.5 \times 4.15 \approx 10.38 \text{ months}$$

- **Team Size**:

$$\text{Team Size} = \frac{58.59}{10.38} \approx 5.64 \approx 6 \text{ persons}$$

3. Embedded Mode

- **Effort**:

$$E = 3.6 \times (15)^{1.20}$$

Compute $15^{1.20}$:

$$15^{1.20} \approx 15 \times 15^{0.20} \approx 15 \times 1.555 = 23.33$$

(Using $15^{0.20} \approx e^{0.20 \times \ln 15} \approx e^{0.20 \times 2.708} \approx e^{0.542} \approx 1.555$).

$$E = 3.6 \times 23.33 \approx 83.99 \text{ person-months}$$

- **Development Time**:

$$T = 2.5 \times (83.99)^{0.32}$$

Compute $83.99^{0.32}$:

$$83.99^{0.32} \approx e^{0.32 \times \ln 83.99} \approx e^{0.32 \times 4.43} \approx e^{1.418} \approx 4.13$$

$$T = 2.5 \times 4.13 \approx 10.33 \text{ months}$$

- **Team Size**:

$$\text{Team Size} = \frac{83.99}{10.33} \approx 8.13 \approx 8 \text{ persons}$$

COCOMO Estimates for Task Management App (15 KLOC)

Development Mode	Effort (Person-Months)	Development Time (Months)	Team Size (Persons)
Organic	40.18	10.18	4
Semi-Detached	58.59	10.38	6
Embedded	83.99	10.33	8

Notes:

- **Effort** is calculated as (E = a \times (KLOC)^b).
- **Development Time** is calculated as (T = c \times (E)^d).
- **Team Size** is calculated as (E / T), rounded to the nearest integer.
- The **Organic** mode is most suitable given the stable requirements and small, experienced team, but estimates for all modes are provided as requested.

Solve the following problems

A gaming company is developing a multiplayer online game with an estimated size of 50 KLOC. The project involves moderate complexity and a team with mixed experience levels. Use the COCOMO model to compute the effort, development time, and team size for all three development modes. How would you adjust the team size if the project deadline is fixed at 12 months?

A logistics company is building a route optimization system estimated at 35 KLOC, requiring integration with GPS and real-time data feeds. Using the COCOMO model, calculate the effort, development time, and team size for all three development modes. If the project must be completed in 10 months, how would you adjust the team size for the most suitable mode?